Hyaline Fibromatosis Syndrome (ANTXR2): Comprehensive Disease Characterization

Disease: Hyaline Fibromatosis Syndrome (HFS) **MONDO:** MONDO:0009229 | **OMIM:** 228600 | **Orphanet:** ORPHA:2176 **Causal gene: ANTXR2** (a.k.a. **CMG2**, capillary morphogenesis gene/protein 2), HGNC:21732, chromosome 4q21.21, UniProt P58335 **Category:** Mendelian, autosomal recessive

---

Summary

Hyaline Fibromatosis Syndrome (HFS) is an ultra-rare autosomal recessive connective-tissue disorder caused by biallelic loss-of-function mutations in the gene ANTXR2 (also called CMG2, capillary morphogenesis gene 2). The disease is defined by progressive deposition of amorphous, PAS-positive, diastase-resistant "hyaline" extracellular matrix (ECM) in skin, gingiva, joints, bone, and—in its severe form—internal viscera. HFS is a unifying term that merges two conditions historically described separately: the milder **Juvenile Hyaline Fibromatosis (JHF)** and the severe, frequently lethal **Infantile Systemic Hyalinosis (ISH)**. These were proven allelic in 2003 when mutations in CMG2/ANTXR2 were identified across families with both presentations.

Mechanistically, ANTXR2 is a type I transmembrane protein that functions as a **collagen VI turnover receptor**. In the healthy state it binds collagen VI (and laminin / collagen IV via its von Willebrand factor A domain) and internalizes it for intracellular degradation through a ligand-triggered "talin-to-RhoA switch." When ANTXR2 is lost, this receptor-mediated endocytic clearance fails, collagen VI accumulates in the extracellular matrix, and the hallmark hyaline material builds up. This model is now firmly supported in patient tissue and in `Antxr2`-null mice, in which the phenotype is rescued by concurrent deletion of `Col6a1`, proving it is collagen-VI-dependent.

Clinically, HFS presents from birth or early infancy with painful joint contractures, generalized skin stiffening with pearly papules and subcutaneous nodules, gingival hypertrophy, osteolytic bone lesions, and—in severe ISH—protein-losing enteropathy, recurrent infections, failure to thrive, and death before age two. Genotype predicts severity: truncating mutations and missense variants in the ectodomain-encoding exons produce severe disease, whereas C-terminal missense variants produce the milder form. No curative therapy exists; management is symptomatic and multidisciplinary, and prevention is limited to genetic counseling and DNA-based prenatal/pre-implantation testing in at-risk (often consanguineous) families.

Key Findings

Finding 1 — HFS is an autosomal recessive connective-tissue disorder caused by biallelic loss-of-function ANTXR2 (CMG2) mutations

HFS unifies two previously separate clinical entities—JHF (milder) and ISH (severe)—shown to be allelic. The landmark study by Hanks et al. (2003) identified **15 different mutations in the gene encoding capillary morphogenesis protein 2 (CMG2/ANTXR2)** on chromosome 4q21 across 17 families with either JHF or ISH. ANTXR2 encodes a transmembrane von Willebrand factor A (vWA)-domain protein that binds laminin and collagen IV/VI and also serves as the cellular receptor for anthrax toxin.

> "We now report the identification of 15 different mutations in the gene encoding capillary morphogenesis protein 2 (CMG2) in 17 families with JHF or ISH. CMG2 is a transmembrane protein that is induced during capillary morphogenesis and that binds laminin and collagen IV via a von Willebrand factor type A (vWA) domain." — PMID: 14508707
>
> "Hyaline fibromatosis syndrome (HFS) is the unifying term for infantile systemic hyalinosis and juvenile hyaline fibromatosis. HFS is a rare autosomal recessive disorder of the connective tissue caused by mutations in the gene for anthrax toxin receptor-2 (ANTXR2)." — PMID: 30176098

Finding 2 — ANTXR2/CMG2 is a collagen VI turnover receptor; its loss causes ECM accumulation via a blocked talin-to-RhoA endocytic switch

CMG2 binds collagen VI (and laminin/collagen IV via its vWA domain) and mediates its endocytic internalization and degradation. Bürgi et al. (2020) showed that CMG2 binds talin—and thereby the actin cytoskeleton—**only in its ligand-free state**; extracellular ligand binding triggers src-dependent talin release and recruitment of RhoA, and this talin-to-RhoA switch is required for intracellular degradation of collagen VI in human fibroblasts.

> "CMG2 binds talin, and thereby the actin cytoskeleton, only in its ligand-free state. Extracellular ligand binding leads to src-dependent talin release and recruitment of the actin cytoskeleton regulator RhoA and its effectors." — PMID: 32428455

The in vivo proof came from Metti et al. (2026), who showed ANTXR2 is expressed predominantly by **fibro-adipogenic progenitors (FAPs)**, and that `Antxr2`-null mice exhibit premature collagen VI accumulation, tissue stiffening, and maladaptive muscle hypertrophy. Crucially, `Antxr2-/-::Col6a1-/-` double-knockout mice rescue the phenotype, demonstrating it is collagen-VI-dependent.

> "ANTXR2-null (Antxr2-/-) mice exhibit premature COL6 accumulation within intramuscular connective tissue, leading to tissue stiffening and progressive, maladaptive muscle hypertrophy with impaired locomotion and myopathic features. Using Antxr2-/-::Col6a1-/- double-knockout mice, we demonstrate that these phenotypes are COL6-dependent." — PMID: 42878719

At the protein level, HFS-causing missense mutations in the vWA / Ig-like ectodomain cause protein misfolding and ER retention (ERAD). Deuquet et al. (2011) demonstrated that such mutations lead to retention of the mutated protein in the endoplasmic reticulum, and that proteasome inhibitors can partially rescue surface expression.

> "missense mutations that map to the extracellular von Willebrand domain or the here characterized Ig-like domain of CMG2 lead to folding defects and thereby to retention of the mutated protein in the endoplasmic reticulum (ER)." — PMID: 21328543

Finding 3 — Genotype–phenotype correlation: ectodomain/truncating mutations → severe ISH; C-terminal missense → milder JHF

In a review of 84 published cases, Casas-Alba et al. (2018) reinforced that **missense mutations in exons 1–12 and mutations leading to a premature stop codon produce the severe (grade 4, life-limiting) form**, while missense pathogenic variants in exons 13–17 produce the mild form. Grade 4 patients present in the newborn period with extreme pain on minimal handling; causes of death are intractable diarrhea, recurrent infections, and organ failure, with a median age of death of 15.0 months (IQR 9.5–24.0).

> "This review provides evidence to reinforce the previous hypothesis that missense mutations in exons 1-12 and mutations leading to a premature stop codon lead to the severe form of the disease, while missense pathogenic variants in exons 13-17 lead to the mild form of the disease." — PMID: 30176098
>
> "Causes of death are intractable diarrhea, recurrent infections, and organ failure. Median age of death of grade 4 cases is 15.0 months (p25-p75: 9.5-24.0)." — PMID: 30176098

Reported pathogenic variants include missense (e.g., c.945T>G p.Cys315Trp; c.652T>C p.Cys218Arg; c.1223T>C p.Leu408Pro), frameshift/truncating (c.1073dup p.Ala359Cysfs*13; c.1074del/c.1074delT), and splice-site (c.697+1G>A) changes, affecting the vWA and Ig-like ectodomain and the cytoplasmic domain. A novel c.1223T>C (p.Leu408Pro) variant appeared to permit a protracted course (P 32196989).

Finding 4 — Core clinical phenotype

Hallmarks across case series are: joint contractures/stiffness (near-universal), generalized skin stiffness/thickening with pearly papules and subcutaneous nodules, hyperpigmented papules over extensor/bony surfaces, fleshy perianal masses, gingival hypertrophy, and severe/persistent diarrhea. Histopathology shows bland fibroblasts embedded in abundant amorphous eosinophilic hyaline matrix that is strongly PAS-positive and diastase-resistant.

> "The hallmarks of the disease are joint contractures, generalized skin stiffness, hyperpigmented papules over extensor surfaces of joints, fleshy perianal masses, severe diarrhea, and gingival hypertrophy." — PMID: 38814306

Severe ISH adds hyaline deposits in viscera, **protein-losing enteropathy** driven by enhanced duodenal collagen deposition, recurrent infections, failure to thrive, osteopenia/osteolytic lesions, bone fractures, and hypotonia.

> *"Enhanced Collagen Deposition in the Duodenum of Patients with Hyaline Fibromatosis Syndrome and Protein Losing Enteropathy."* — PMID: 33147779

Rare presentations include hypercalcemia (from osteoclast stimulation and immobilization; P 37264371) and, uncommonly, isolated gingival enlargement as the only manifestation (P 34627224).

Suggested HPO terms: HP:0003549 (abnormal connective tissue), HP:0001371 (flexion contracture), HP:0000704 (gingival overgrowth), HP:0000979 (purpura/skin papules — approximate), HP:0002664→HP:0100242 (subcutaneous nodules), HP:0002754 (osteolysis), HP:0002028 (chronic diarrhea), HP:0001824 (weight loss/failure to thrive), HP:0002902 (hyponatremia—secondary), HP:0003072 (hypercalcemia).

Finding 5 — Epidemiology: ultra-rare, consanguinity-associated, recurrent/ founder-type mutations

HFS is ultra-rare. Orphanet lists it as ORPHA:2176 with prevalence **<1/1,000,000** and fewer than ~100–150 reported cases worldwide (Casas-Alba 2018 reviewed 84 cases; Futagawa 2024 noted ~100 cases reported). It is strongly associated with consanguineous backgrounds, and over 40 ANTXR2 mutations are known.

> "Over 40 mutations in the ANTXR2 gene have been associated with cases of HFS." — PMID: 29801470

The recurrent c.1073dup (p.Ala359Cysfs*13) mutation appears to arise sporadically/recurrently in different populations rather than from a single ancient founder:

> "Haplotype analysis was performed to determine the origin of this mutation in this consanguineous cohort, which suggested that it may develop sporadically in different populations." — PMID: 29801470
>
> "Thus far, about 100 cases have been reported but few of these were reported from Japan." — PMID: 39258861

Inheritance is autosomal recessive (homozygous or compound heterozygous). No sex predilection is established, and intellectual ability is generally normal.

Finding 6 — Diagnosis: characteristic biopsy + ANTXR2 molecular confirmation; imaging shows osteolytic/intra-articular lesions

Diagnosis is clinical + histopathological + molecular. Skin/nodule biopsy shows bland spindle/epithelioid fibroblasts embedded in abundant amorphous eosinophilic hyaline ground substance that is PAS-positive and diastase-resistant. Confirmation is by ANTXR2 sequencing — single-gene testing, trio whole-exome sequencing, or rapid whole-genome sequencing.

> "rapid whole-genome sequencing (rWGS) was done with Fabric GEM®-assisted artificial intelligence." — PMID: 38989346

Radiographic/MRI findings include osteolytic/low-density bone lesions, osteopenia, and multifocal intra-articular hyaline deposition.

> "We describe a case of a 25-year-old patient presenting with bilateral knee pain, limited range of motion in her extremities, and lower extremity weakness with detailed MR imaging demonstrating the first case of multifocal intra-articular deposition of hyaline material within several joints." — PMID: 33120290

Differential diagnosis includes juvenile rheumatoid arthritis, Winchester syndrome, lipoid proteinosis, Farber disease, neurofibromatosis, and gingival fibromatosis.

Finding 7 — No curative therapy; management is symptomatic and multidisciplinary

No specific or curative treatment exists. Management is supportive and multidisciplinary: surgical excision of debilitating/giant nodules (often with recurrence), intralesional/oral corticosteroids (temporary nodule reduction), analgesia for chronic pain, nutritional support for failure to thrive/protein-losing enteropathy, physiotherapy for contractures, and periodontal/dental surgery for gingival hypertrophy (improves mastication, weight gain, and quality of life).

> "No specific treatment is yet available. Most patients with the severe form of the condition pass away before the second year of age." — PMID: 38814306
>
> "Early surgical treatment provided instantaneous and more sustainable results, while corticosteroids can be used as alternative modalities with temporary outcomes." — PMID: 33173631

Mechanistically-motivated experimental strategies include proteasome inhibitors to rescue ER-retained misfolded CMG2 mutants in vitro (P 21328543) and S-acylation / APT2-pathway modulation (P 42409807) — none yet in clinical use.

Suggested NCIT terms: Corticosteroid Therapy, Surgical Excision, Physical Therapy, Nutritional Support, Analgesic Therapy.

Finding 8 — Antxr2/Cmg2 knockout mice model multiple HFS mechanisms

The `Antxr2-/-` mouse (*Mus musculus*, NCBI Taxon 10090; ortholog Antxr2, NCBI Gene 71914) is the principal model and recapitulates several facets of HFS:

1. **ECM/collagen VI accumulation → muscle hypertrophy**, rescued by `Col6a1` double-KO (P 42878719).

2. **Impaired intestinal stem-cell renewal after colitis** via a defective fetal-to-Lgr5^{+} transition and β-catenin nuclear translocation — identifying CMG2 as a context-specific Wnt modulator and offering a mechanism for lethal diarrhea in HFS.

> "CMG2 knockout impairs the transition from fetal-like to Lgr5+ adult stem cells, which is associated with a defect in ß-catenin nuclear translocation. Based on our findings, we propose that CMG2 functions as a context-specific modulator of Wnt signaling, essential for replenishing the pool of intestinal stem cells following injury." — PMID: 40847216

1. **Cervical ECM/collagen accumulation → parturition (dystocia) defect** with altered pyridinoline crosslink density.

> "Antxr2-/- females are known to have a parturition defect, in part, due to an excessive accumulation of extracellular matrix proteins in the cervix, particularly collagen." — PMID: 24390076

1. **Angiogenesis/chemotaxis role**: CMG2 is required for growth-factor-induced endothelial chemotaxis in a Rho-dependent manner.

> "CMG2 mediates the antiangiogenic effects of PA and is required for growth-factor-induced chemotaxis." — PMID: 35212873

1. **Anthrax resistance**: `Antxr2`-knockout mice are resistant to anthrax toxin / *Bacillus anthracis* (reflecting CMG2's dual role as the major anthrax toxin receptor). Zebrafish models additionally show CMG2 controls oriented cell division.

Finding 9 — Identifiers, nomenclature, and gene/protein annotations

Resource	Identifier
MONDO	MONDO:0009229
OMIM	228600 (Hyaline Fibromatosis Syndrome; formerly Systemic Hyalinosis)
Orphanet	ORPHA:2176
MeSH	"Hyalinosis, Systemic"
ICD-10 (proxy)	L98.5 (mucinosis/hyalinosis of skin) or M72.x (fibromatoses)
Gene	**ANTXR2** (HGNC:21732; NCBI Gene 118429; Ensembl ENSG00000163297), chr 4q21.21; aliases CMG2
Protein	UniProt **P58335** (ANTR2_HUMAN)

Historical synonyms: Juvenile Hyaline Fibromatosis (JHF), Infantile Systemic Hyalinosis (ISH; formerly OMIM 236490), Murray-Puretic-Drescher syndrome, fibromatosis hyalinica multiplex juvenilis, systemic hyalinosis, molluscum fibrosum. The ANTXR2 protein is a type I single-pass transmembrane protein with an extracellular vWA/I-domain (containing a MIDAS metal-ion-dependent adhesion site), an Ig-like domain, a transmembrane helix, and a cytoplasmic tail.

> "Hyaline fibromatosis syndrome (HFS; MIM 228600) is a rare autosomal recessive disorder." — PMID: 30152846
>
> "we previously mapped the gene for JHF to chromosome 4q21." — PMID: 14508707

Information is derived from aggregated disease-level resources (OMIM, Orphanet) combined with individual case reports/series (EHR-type data).

Finding 10 — Temporal course and prevention

Onset ranges from congenital to early infancy. Severe ISH usually presents at or shortly after birth (extreme pain on handling, stiffness) and is progressive, with death typically within the first 1–2 years. Milder JHF presents in infancy/early childhood with subcutaneous nodules that increase in number and size with age and can continue into adulthood (reported age range 5–44 years).

> "The condition usually presents at birth, resulting in death in infancy." — PMID: 22300424
>
> "With increasing age, the number and size of the nodules gradually increased in the patient." — PMID: 30152846

The course is chronic, progressive, and non-remitting; no spontaneous remission occurs, and nodules recur after excision. No true primary prevention exists for a monogenic recessive disorder. Reproductive prevention is feasible: identification of the familial ANTXR2 mutation enables carrier testing, cascade screening, presymptomatic diagnosis, DNA-based prenatal diagnosis, and preimplantation genetic testing, with genetic counseling for consanguineous couples (25% recurrence risk per pregnancy).

> "identification of pathogenetic mutations in the ANTXR2 gene makes DNA-based prenatal diagnosis feasible for subsequent pregnancies." — PMID: 22300424

Finding 11 — Collagen VI is the physiological ligand that accumulates in HFS tissue

Bürgi et al. (2017) demonstrated that CMG2/ANTXR2 is a central regulator of collagen VI homeostasis: CMG2 loss of function promotes collagen VI accumulation in patients, driving nodule formation, with massive collagen VI accumulation in `Antxr2`-null mouse uteri. Abrami et al. (2026) states plainly that CMG2/ANTXR2 functions as a collagen VI receptor required for ECM homeostasis.

> "CMG2 loss of function promotes accumulation of collagen VI in patients, leading in particular to nodule formation." — PMID: 28604699
>
> "CMG2/ANTXR2 functions as a Collagen VI receptor required for extracellular matrix homeostasis." — PMID: 42409807

This identifies the specific "hyaline" ECM material as collagen VI, bridging the original observation that CMG2 binds laminin/collagen IV (Hanks 2003) with the modern collagen-VI turnover model.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function mutation in *ANTXR2* (CMG2)** — missense (ectodomain), frameshift, nonsense, or splice-site — *leads to* loss of functional cell-surface receptor.
2. *Branch (ectodomain missense):* vWA/Ig-domain misfolding *results in* ER retention and ERAD-mediated degradation, so little/no mature protein reaches the surface (demonstrated in vitro;

   P 21328543

).
3. *Branch (truncating/splice):* loss of protein entirely → severe ISH.
4. **Loss of surface ANTXR2 on fibro-adipogenic progenitors and fibroblasts** *results in* failure of the ligand-triggered talin-to-RhoA switch that drives receptor-mediated endocytosis (demonstrated in human fibroblasts;

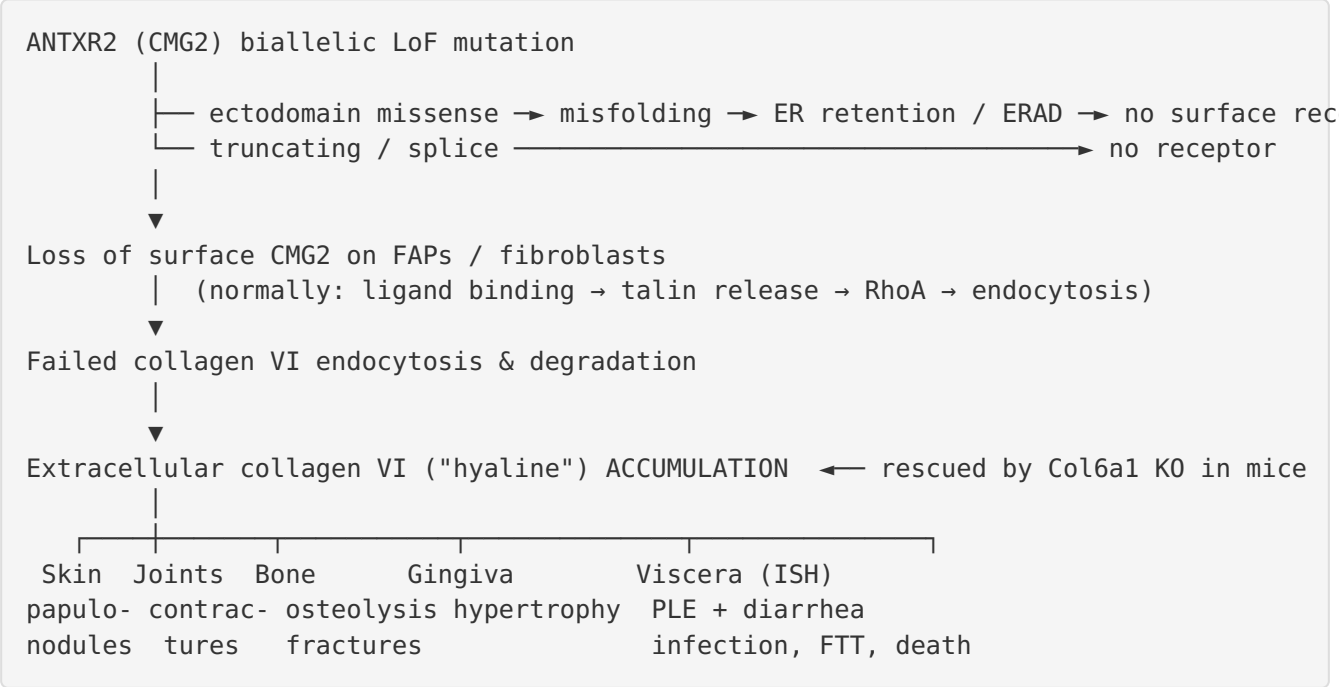
   P 32428455

).
5. **Failed endocytic internalization/degradation of collagen VI** *leads to* progressive extracellular accumulation of collagen VI and associated matrix (demonstrated in patient tissue and mouse; PMIDs 28604699, 42878719).
6. **Accumulated collagen VI / "hyaline" ECM** (PAS-positive, diastase-resistant) *results in* tissue stiffening and deposition in skin, gingiva, perichondrium/joints, bone, and viscera.
7. **Tissue-level deposition** *leads to* the clinical hallmarks:
8. Skin/subcutis → papulonodules, pearly papules, perianal masses.
9. Joints/periarticular → painful contractures, stiffness, limited range of motion.
10. Bone → osteolytic lesions, osteopenia, fractures.
11. Gingiva → hypertrophy.
12. *Branch (severe ISH):* duodenal/intestinal collagen deposition *leads to* protein-losing enteropathy; impaired intestinal stem-cell renewal (Wnt/β-catenin defect, *inferred* from mouse colitis model;

    P 40847216

) *contributes to* intractable diarrhea → failure to thrive, infection susceptibility, and early death.

Schematic



Upstream vs downstream; cell types and processes

- **Upstream (initiating):** ANTXR2 loss of function; ER quality-control failure of misfolded mutants.
- **Central node:** defective collagen VI receptor-mediated endocytosis (talin→RhoA switch).
- **Downstream (effector):** extracellular collagen VI accumulation and consequent tissue fibrosis/stiffening.
- **Key cell types:** fibro-adipogenic progenitors (FAPs), dermal and gingival fibroblasts, intestinal stem cells (Lgr5^{+}), endothelial cells (angiogenesis role).
- **Suggested GO terms:** GO:0030198 extracellular matrix organization; GO:0006898 receptor-mediated endocytosis; GO:0030574 collagen catabolic process; GO:0016055 Wnt signaling pathway; GO:0001525 angiogenesis; GO:0030968 ER unfolded protein response.
- **Suggested CL terms:** CL:0000057 (fibroblast), CL:0000134 (mesenchymal stem cell), CL:0000115 (endothelial cell), CL:0002254 (intestinal stem cell).
- **Suggested UBERON terms:** UBERON:0002097 (skin), UBERON:0001828 (gingiva), UBERON:0000982 (joint), UBERON:0002481 (bone tissue), UBERON:0002114 (duodenum), UBERON:0000002 (uterine cervix — model).

- **Relevant molecule (CHEBI/complex):** collagen type VI (ECM protein complex); PAS-reactive hyaline glycoprotein material.

Evidence Base

PMID	Title (abbrev.)	Role in this report
14508707	CMG2 mutations cause JHF and ISH	Establishes ANTXR2/CMG2 as the causal gene; JHF+ISH allelic; 4q21 locus
30176098	HFS clinical update & genotype-phenotype	Unifying HFS term; exon-based genotype–phenotype rule; mortality data
32428455	Collagen VI receptor talin-to-RhoA switch	Defines endocytic molecular switch controlling collagen VI degradation
42878719	FAPs / ANTXR2 fine-tune collagen VI	In vivo COL6-dependence; `Col6a1` double-KO rescue
21328543	Ectodomain mutations rescued by proteasome inhibitors	ER retention of misfolded mutants; therapeutic concept
28604699	CMG2/ANTXR2 regulates extracellular collagen VI	Collagen VI accumulation in patients
42409807	Dynamic S-acylation controls CMG2	Confirms CMG2 as collagen VI receptor; S-acylation pathway
38814306	HFS: a recognizable syndrome	Core hallmarks; no specific treatment; prognosis
33147779	Duodenal collagen & protein-losing enteropathy	Mechanism of PLE in severe HFS
29801470	Genetic basis in consanguineous cohort	>40 mutations; recurrent-mutation origin
39258861	Prognostic factors for wellbeing	~100 cases reported globally
33120290	Multifocal intra-articular involvement	MRI hyaline deposition imaging
38989346	AI-assisted rapid WGS diagnosis	rWGS diagnostic approach
33173631	JHF treated with excision + corticosteroid	Symptomatic treatment outcomes
40847216	Injury-induced intestinal stem-cell renewal needs CMG2	Wnt/β-catenin mechanism for lethal diarrhea
24390076		Mouse ECM accumulation phenotype

PMID	Title (abbrev.)	Role in this report
	Cervical collagen remodeling / parturition defect	
35212873	CMG2 and growth-factor-induced angiogenesis	Angiogenesis/chemotaxis role
22300424	ISH with splice-site ANTXR2 mutation	Congenital onset; prenatal diagnosis feasibility
30152846	Two novel ANTXR2 mutations (Chinese)	MIM 228600; progressive nodule course
37264371	Hypercalcemia presentation	Rare metabolic complication
34627224	Gingival enlargement as sole manifestation	Phenotypic variability; diagnostic pitfall
25781883	N-glycosylation of CMG2/TEM8	Glycosylation buffers against pathogenic mutations

The evidence base spans **human clinical case series and reviews** (diagnosis, phenotype, genotype–phenotype, epidemiology, treatment), **in vitro mechanistic studies** in patient/human fibroblasts (endocytic switch, ER retention), and **model-organism studies** (Antxr2 knockout mouse; zebrafish), providing convergent support for the collagen-VI-turnover model. The most decisive single piece of evidence is the `Antxr2⁻/⁻::Col6a1⁻/⁻` genetic rescue (P 42878719), which moves collagen VI from a correlative accumulant to a causal effector.

Limitations and Knowledge Gaps

1. **Rarity limits quantitative epidemiology.** With <1/1,000,000 prevalence and ~100 reported cases, there are no robust incidence, survival-curve, or population-stratified data; most figures derive from case reviews (e.g., the 84-case review) rather than registries.
2. **Phenotype frequencies are semi-quantitative.** Precise penetrance of individual HPO features (e.g., exact % with osteolytic lesions vs. contractures) is not established from systematic cohorts.

3. **Mechanistic gaps downstream of collagen VI.** How collagen VI accumulation specifically produces bone osteolysis, hypercalcemia, and the striking pain phenotype is not fully resolved. The Wnt/stem-cell mechanism for diarrhea is demonstrated in a mouse colitis model and remains *inferred* for human PLE.
4. **Therapeutics are preclinical.** Proteasome-inhibitor rescue of ER-retained mutants and APT2/S-acylation modulation are in vitro only; no genotype-tailored (e.g., mutation-class-specific) therapy has reached trials.
5. **No approved biomarker** tracks disease activity; diagnosis still relies on biopsy + sequencing rather than a circulating marker.
6. **Modifier genes and expressivity** within the same genotype are not characterized; why some ectodomain variants permit protracted courses (e.g., p.Leu408Pro) is incompletely understood. N-glycosylation state may buffer tolerance to mutations (P 25781883).

Proposed Follow-up Experiments / Actions

1. **Natural-history registry.** Establish an international HFS registry capturing standardized HPO annotations, genotype, and longitudinal outcomes to produce real penetrance, progression-rate, and survival statistics.
2. **Mutation-class-specific therapeutic testing.** For ectodomain misfolding mutants, test proteasome inhibitors / chemical chaperones / ERAD modulators in patient-derived iPSC fibroblasts and FAP organoids to see whether surface CMG2 and collagen VI clearance are restored; progress toward mutation-stratified trials.
3. **Collagen-VI-lowering strategies.** Given the `Col6a1` genetic rescue, evaluate COL6-targeting approaches (antisense oligonucleotides / siRNA against COL6A1-3) in `Antxr2-/-` mice as a potential disease-modifying therapy.
4. **Biomarker development.** Measure circulating collagen VI fragments (e.g., C6M/PRO-C6 neoepitopes) in HFS patients vs. controls as candidate diagnostic/monitoring biomarkers.
5. **Dissect the diarrhea/PLE mechanism in humans.** Combine duodenal biopsy transcriptomics/single-cell analysis with Wnt/β-catenin readouts to test whether the mouse intestinal stem-cell-renewal defect operates in human ISH.

6. **Pain and bone mechanisms.** Investigate how periarticular/osseous collagen VI deposition drives nociception and osteolysis (osteoclast assays, immobilization-hypercalcemia models) to identify adjunctive symptomatic targets.
7. **Genotype–modifier studies.** Sequence modifier loci and N-glycosylation determinants across patients with discordant severity for the same primary variant.

Conclusion

Hyaline Fibromatosis Syndrome is a Mendelian, autosomal recessive ECM disorder in which **loss of the collagen VI receptor ANTXR2/CMG2** abolishes receptor-mediated collagen VI endocytosis and degradation, causing progressive hyaline (collagen VI) accumulation across connective tissues. The molecular chain—from biallelic loss-of-function mutation, through failed talin-to-RhoA endocytic switching, to collagen-VI-dependent tissue stiffening—is now supported by patient tissue, human fibroblast biology, and a decisive `Col6a1` double-knockout mouse rescue. Clinical severity tracks mutation type/position, diagnosis combines characteristic PAS-positive hyaline histology with ANTXR2 sequencing, and—absent any curative therapy—care remains symptomatic and multidisciplinary, with prevention resting on genetic counseling and prenatal/pre-implantation testing in at-risk, frequently consanguineous families.
